

VATSAL S. SUTHAR

Ahmedabad, Gujarat, India | +91-8401571819 | vatsal97.ss@gmail.com
linkedin.com/in/vatsal-suthar-476587136 | github.com/vatsalsuthar

SUMMARY

Data Science and Analytics professional with extensive experience in data analysis, machine learning, statistical modeling, and BI-driven consulting. Proven track record of designing end-to-end analytics solutions, building production-grade machine learning systems, and delivering measurable business impact through data-driven decision-making. Skilled in Python, SQL, Power BI, and cloud-based deployment of ML models.

TECHNICAL SKILLS

Programming Languages: Python, R

Machine Learning & AI: Scikit-Learn, XGBoost, SMOTE, Optuna, Model Deployment (MLOps)

Databases: MySQL, MongoDB, DynamoDB, SQL

BI & Analytics: Power BI, Tableau, Excel, Google Analytics

Clouds & Frameworks: AWS (EC2, FastAPI), GCP, Docker

Tools & IDE: PyCharm, VS Code, Spyder, Google Colab, Jupyter

Soft Skills: Time Management, Teamwork, Problem Solving, Research Skills, Engaging Presentation

PROFESSIONAL EXPERIENCE

Lead Data & Analytics Consultant (Project-Based)

Sep 2024 - May 2026

Remote / Independent

Enterprise Supply Chain Analytics Dashboard (Power BI / Data Analyst Project)

- Designed and deployed an end-to-end Power BI executive dashboard tracking supply chain KPIs, reducing inventory overhead costs by 14% and identifying \$120K in annual logistics leakage.
- Engineered robust ETL pipelines using Power Query and SQL to ingest, clean, and transform 5M+ rows of multi-source ERP data, maintaining a data refresh latency of under 15 minutes.
- Developed complex DAX measures and advanced data models (Star Schema) to deliver dynamic, interactive views on vendor performance, fulfillment cycles, and demand forecasting.

Predictive Customer Churn & Personalization Engine (AI/ML Project)

- Built a scalable, production-grade customer churn prediction system for an e-commerce platform using Python and XGBoost, increasing customer retention by 18%.
- Developed the entire ML lifecycle (MLOps) using Scikit-Learn for advanced feature engineering, resolving extreme class imbalance with SMOTE, and tuning hyperparameters via Optuna to reach a 94.2% ROC-AUC score.
- Containerized the inference pipeline using Docker and deployed it as a REST API on AWS (EC2/FastAPI), enabling automated weekly batch-scoring to sync with marketing automation workflows.

Research Associate - IIM Ahmedabad

June 2022 - March 2024

- Led an NLP project predicting Twitter users' gender and age using tokenization, TF-IDF, and reference dictionaries, achieving high accuracy.
- Contributed to a research project analyzing MSME survey data across regions to identify trends and insights, supporting policy decisions and MSME growth.
- Collaborated with a professor to create an online course on digital tech applications, boosting engagement by 30% through effective Power BI data visualization.

Research Machine Learning Intern - ISRO Ahmedabad

June 2021 - March 2022

- Gained insights into SAR data and researched the Edge-Otsu algorithm for detecting water and land features, improving accuracy by 20% in water mapping.
- Leveraged ESA SNAP tools for comprehensive data processing and analysis.

PROJECTS

Human Activity Recognition Using Smartphones

- Utilized various machine learning algorithms, including K-Nearest Neighbors (KNN), Support Vector Classification (SVC), and Random Forest, to enhance model performance; achieved the highest accuracy of 96.19% using Logistic Regression.

NYC Yellow Taxi Prediction Analysis

- Analyzed NYC taxi attribute data to predict fares with 93.01% accuracy using the XGBoost regression model, highlighting strong predictive capabilities.

Covid-19 Chest X-ray Detection

- Developed a robust model using ResNet18 for evaluating chest X-rays; successfully deployed the model on the cloud, enabling real-time analysis and accessibility for healthcare professionals.

EDUCATION

M.Tech. in Computer Science and Engineering

Sep 2020 - May 2022

Ahmedabad University - Specialization in Data Science and Analytics | CGPA: 3.00/4.00

Bachelor of Engineering in Computer Engineering

Aug 2015 - June 2019

Gujarat Technological University - Darshan University | CGPA: 8.21/10.00

CERTIFICATES

- R for Data Science - applied R techniques for exploratory data analysis and visualization
- Cloud Foundations - gained foundational knowledge of cloud computing principles
- Python Fundamentals for Beginners by Great Learning - developed a strong foundation in Python programming
- Career Essentials in Data Analysis by Microsoft and LinkedIn - acquired essential skills in data analysis